

System and evidence

Five-second vessel classification from public DeepShip recordings

OPERATING ASSUMPTION A near-sensor processor receives hydrophone audio, converts each non-overlapping 5 s interval to one four-class prediction, and may retain scores locally. Hydrophone, ADC and field hardware are not yet measured.

Implemented signal and model path



Feature baseline versus deployment CNN

Model	Accuracy	Macro F1	Tug recall
Logistic regression / 73 handcrafted features	0.393	0.283	0.025
Selected CNN / FP32 ONNX deployment candidate	0.067	0.031	0.000

EVIDENCE CHECK The strongest test result remains logistic regression at macro F1 0.283; the final CNN reached 0.031. Fast edge execution does not establish classification readiness.

Test confusion: where the models fail

LOGISTIC REGRESSION

		predicted			
		C	P	Ta	Tu
true	C	3	0	34	0
	P	4	40	1	0
	Ta	1	3	9	0
	Tu	0	30	9	1

counts; n = 135 windows

SELECTED CNN / FP32 ONNX

		predicted			
		C	P	Ta	Tu
true	C	0	0	37	0
	P	0	0	45	0
	Ta	4	0	9	0
	Tu	1	0	39	0

counts; n = 135 windows

Development-machine latency, memory and size

Path / boundary	median	p99	model	memory
Logistic feature extraction	7.18 ms	not measured	-	not measured
Logistic inference	0.82 ms	not measured	7,945 B	not measured
Logistic end to end	8.13 ms	not measured	-	not measured
ONNX DSP / inference	3.67 / 0.88 ms	- / 1.07 ms	116,187 B	+39.8 MiB*
ONNX full warm-cache pipeline	6.70 ms	13.19 ms	23,668 params	333.6 MiB peak*

* Python process + ONNX Runtime + DSP libraries; not model-only memory.

Edge-first design and risks

Proposed deployment path; only preprocessing and inference are implemented

Proposed near-sensor architecture



WHY INFER NEAR THE SENSOR

Avoid continuous audio uplink, bound response time, tolerate intermittent connectivity and store only events or review clips. This benefit depends on calibrated false-alarm behavior and target-device power.

COMPUTE / POWER / STORAGE / LINK

FP32 remains recommended: 113.5 KiB and 13.2 ms warm-cache p99 on Apple M2. Dynamic INT8 saved 1.8% but was 6.4% slower at p99. Static INT8 saved 55.4% but lost 0.025 macro F1 and 0.111 maximum recall.

Decision logic that is not deployment-calibrated

THRESHOLD + TEMPORAL AGGREGATION

Retain the four per-window scores. Calibrate class thresholds on representative field data, then require evidence across multiple windows before an event. No production threshold, K-of-N rule or smoothing horizon has been validated.

RARE / UNKNOWN EVENTS

Tug has 118 windows but only 3 vessel groups. A validation-selected Tug threshold (0.195) produced test recall 0.025. No unknown/background set exists, so confidence and embedding distance do not validate unknown rejection.

Risk register: do not treat all errors as model errors

RISK	CURRENT SIGNAL	DEPLOYMENT CONSEQUENCE
Model error	CNN test macro F1 0.031	Misclassification is the observed default.
Label ambiguity	48 manual audits pending	Cannot yet separate bad labels from errors.
Domain shift	Few independent vessels / sites	Field spectra may differ from public subset.
False alarms	No long-duration background run	Precision at scale is unknown.
Site shortcuts	Confidence/distance are proxies	Recording context may drive predictions.

Required before real deployment

- ☐ Representative hydrophones, ADCs, sites and operating conditions
- ☐ Long-duration false-alarm and missed-event evaluation
- ☐ Unknown/background and overlapping-source collection
- ☐ Threshold, smoothing and event-policy calibration
- ☐ Complete 48 label audits and 20 error-listening reviews
- ☐ Target-device power, thermal, latency and RSS benchmarks
- ☐ Independent vessel-disjoint holdout with no prior test reuse
- ☐ Field, enclosure, storage and connectivity validation

BOTTOM LINE

The pipeline is computationally feasible on a laptop CPU. The evidence is not yet sufficient for a safety-, monitoring- or operations-critical underwater deployment.